EE4-14
EE9-SO16

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected Copy

SPEECH PROCESSING

Thursday, 16 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : W. Dai

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SPEECH PROCESSING

1. A spectrogram is defined at time-frame m and frequency index k by

$$X(k,m) = \sum_{r=0}^{N-1} w(r)x(m-r)\exp\left(-\frac{j2\pi k(m-r)}{N}\right)$$

where the input signal is

$$x(n) = \begin{cases} 0 & \text{for } n < 0 \\ \exp\left(\frac{j2\pi fn}{N}\right) & \text{for } n \geq 0 \end{cases}$$

and the window function with length $R \leq N$ is

$$w(r) = \begin{cases} \frac{1}{R} & \text{for } r < R \\ 0 & \text{for } r \geq R. \end{cases}$$

a) i) Draw a fully labelled sketch of the window function. [2]

ii) Consider denoting the upper limit on the summation in the above equation for the spectrogram as $Q-1$. Determine Q for which the terms of the summation are non-zero. [2]

iii) For the case of $0 \leq m < R$, show that [4]

$$|X(k,m)| = \left|\frac{\sin\left(\frac{\pi(f-k)(m+1)}{N}\right)}{R\sin\left(\frac{\pi(f-k)}{N}\right)}\right|.$$

iv) For the case of $m \geq R$, find an expression for $|X(k,m)|$. [2]

The following identities can be used:

$$\sum_{r=0}^{N-1} \exp(j2\pi(ar+b)) = \frac{\sin(N\pi a)}{\sin(\pi a)}\exp(j\pi(a(N-1)+2b)),$$

$$\frac{\sin(ax)}{\sin(bx)} \xrightarrow{x\to 0} \frac{a}{b}.$$

b) Consider the case of $k = f$ and varying m, with $N = 128$. [4]

i) Determine expressions for $|X(k,m)|$ for the ranges of m:

- $m < 0$
- $0 \leq m < R$
- $m \geq R$.

ii) Draw labelled sketch graphs of $|X(k,m)|$ as a function of m for $R = 16$ and $R = 32$. Include both graphs on the same axes and cover the range $-16 \leq m \leq 48$.

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c) Consider the case of $m = 127$ and varying k, with $N = 128$. [3]

Draw labelled sketch graphs of $|X(k,m)|$ as a function of k for $R = 16$ and $R = 32$. Include both graphs on the same axes and cover the range $f - 12 \leq k \leq f + 12$

d) Explain how the choice of R in spectrographic analysis affects the time and frequency resolution. Relate your explanation to your answers in parts (b) and (c). [3]

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2. a) Consider a speech recognizer based on a left-to-right, no-skips Hidden Markov Model. Consider a particular state of this recognizer. The probability of remaining in the state from one frame to the next frame is $(1-p)$.

i) Explain why the state sequence is described as a Markov process. [1]

ii) Prove that the number of frames, D, aligned to this state has a negative exponential distribution with an average of $\frac{1}{p}$. [5]

b) Define and explain the Fisher Ratio and its relevance to speech recognition. Include in your answer [4]

i) a conceptual definition with any appropriate diagrams,

ii) a straightforward mathematical definition,

iii) a definition in terms of appropriately defined matrices,

iv) an explanation of its relevance to speech recognition.

c) In a particular speaker recognition system, a statistical model is trained for each speaker. In the model, the probability density of the N-dimensional input feature vector, $\mathbf{x}$, is modelled by the multivariate Gaussian distribution given by

$$p(\mathbf{x}) = (2\pi)^{-N/2}|\mathbf{C}|^{-1/2}\exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m})^T\mathbf{C}^{-1}(\mathbf{x}-\mathbf{m})\right).$$

The mean $\mathbf{m}$ is defined as $\mathbf{m} = E\{\mathbf{x}\}$ and the covariance matrix $\mathbf{C}$ is given by

$$\mathbf{C} = E\{\mathbf{x}\mathbf{x}^T\} - \mathbf{m}\mathbf{m}^T$$

where $E\{\}$ denotes the expectation operator.

i) Consider such a speaker recognition system using $N = 2$ with models for 2 speakers, speaker A and speaker B. The mean feature vectors for speakers A and B are

$$\begin{bmatrix}2\\3\end{bmatrix} \quad \text{and} \quad \begin{bmatrix}5\\10\end{bmatrix}$$

respectively, and the covariance matrices for both speaker models are identity matrices, $\mathbf{I}$.

Show that, if $\mathbf{x} = \begin{bmatrix}x_1\\x_2\end{bmatrix}$, then the probability that x_1 is less that a threshold T for speaker A is given by

$$pr(x_1 < T) = \Phi(T-2)$$

$$\text{where} \quad \Phi(y) = (2\pi)^{-1/2}\int_{t=-\infty}^{y}\exp\left(-\frac{1}{2}t^2\right)dt.$$

[2]

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ii) The speaker identification system as specified in c) i) determines a test speaker to be speaker A if $x_1 < T$ and to be speaker B if $x_1 \geq T$. Find the probability of correct identification when $T = 2.5$, $T = 3$ and $T = 3.5$. [4]

It can be assumed that the prior probabilities of both speakers is 0.5. Values of the function $\Phi(T)$ can be read from Table 1.

T	0	0.5	1.0	1.5	2.0	2.5
$\Phi(T)$	0.5	0.6915	0.8413	0.9332	0.9772	0.9938

Table 1

d) A linear transformation, $\mathbf{F}$, is applied to the input feature vectors in the system of part c) to give corresponding transformed feature vectors

$$\mathbf{y} = \mathbf{F}\mathbf{x}.$$

i) Explain briefly why it can be advantageous in a speech or speaker recognition system to apply such a transformation, $\mathbf{F}$, to the input feature vectors before they are used as the input of a pattern matching process. [2]

ii) Derive expressions for the mean vector $\mathbf{m}_y$ and the covariance matrix $\mathbf{C}_y$ of $\mathbf{y}$, in terms of $\mathbf{m}$ and $\mathbf{C}$. [2]

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3. a) Let $V(z)$ be the system function representing a vocal tract in the z domain.

i) Write down an expression for the corresponding complex cepstrum coefficients, $c(n)$, in terms of V. [2]

ii) Write down an expression for the z transform of the complex cepstrum coefficients, $C(z)$ in terms of $c(n)$. [1]

iii) Consider the discrete-time sequence

$$x(n) = \delta(n) + \beta\delta(n-P)$$

for P being an integer constant and $|\beta| < 1$. Find the complex cepstrum of $x(n)$. [4]

You may use the power series expansion

$$\log(1-q) = -\sum_{n=1}^{\infty} \frac{q^n}{n} \quad \text{for } |q| < 1.$$

b) Consider the case of the vocal tract filter comprising an all-pole filter with order p

$$V(z) = \frac{1}{A(z)}.$$

Write down general expressions for $A(z)$ in terms of the following parameters. State any assumptions made.

i) Reflection coefficients, r_k. [2]

ii) Log area ratios, g_k. [2]

c) i) Explain the differences between covariance LPC and autocorrelation LPC and state their relative advantages. [3]

ii) A particular signal is made from the sum three tones at 800 Hz, 1 kHz and 1.2 kHz, with relative amplitudes 1, 1 and 0.2. White noise with low relative level is added to this signal. This signal is analysed using 10th order autocorrelation and covariance LPC. In no particular order, Figure 3.1 shows the resulting filter spectra, as functions of normalised frequency.

Determine the relative amplitude in dB of the tone at 1.2 kHz. [1]

Explain what characteristics of these spectra can be used to determine which corresponds to autocorrelation LPC and which corresponds to covariance LPC. [2]

Give an explanation for the high frequency peak observed in the Fig. 3.1(a). [1]

How would you modify the analysis in order to obtain a more accurate spectrum. [2]

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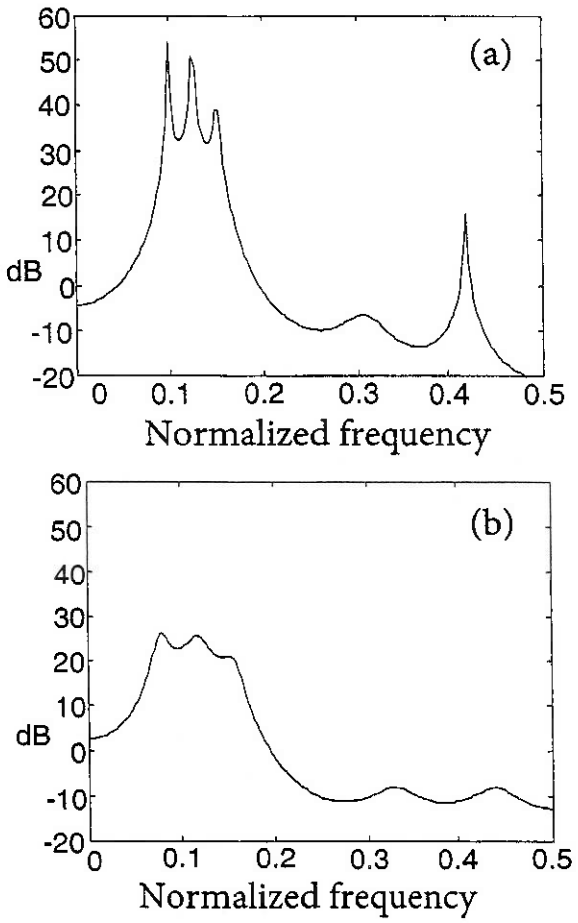


Figure 3.1: Filter spectra

4. a) Define and explain 3 measures for the objective assessment of the quality of speech in a speech coding application. [6]

b) Consider a speech signal $s(n)$ and the block diagram of the main elements of a speech coder shown in Fig. 4.1.

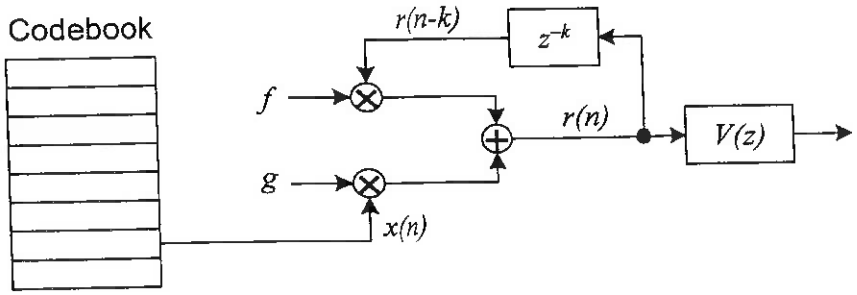


Figure 4.1: Components of a speech coder

i) Identify and explain the operation of the parts of this coder that are responsible for modelling the periodic components and the non-periodic components of the excitation signal $r(n)$. [2]

ii) In this coder, it is decided to determine the values of k and f so as to minimize the energy function

$$E(k,f) = \sum_{n=0}^{N-1} (r(n) - r(n-k)f)^2 .$$

Assume that the value of k is given and $\hat{}$ indicates the optimal value. Show that the optimum values $\hat{f}$ and the corresponding energy function $\hat{E}$ are given by

$$\hat{f}(k) = \frac{\sum_{n=0}^{N-1} r(n)r(n-k)}{\sum_{n=0}^{N-1} r^2(n-k)}$$

and

$$\hat{E}(k) = \sum_{n=0}^{N-1} r^2(n) - \frac{\left(\sum_{n=0}^{N-1} r(n)r(n-k)\right)^2}{\sum_{n=0}^{N-1} r^2(n-k)}.$$

[6]

iii) In a particular operation of this speech coder, the signal $r(n)$ has the following values and is zero for all other n.

$$\begin{aligned} r(-100) &= 9 \\ r(-95) &= 4 \\ r(-50) &= 10 \\ r(-45) &= 7 \\ r(0) &= 5 \\ r(5) &= 8 \end{aligned}$$

For $N = 40$, determine $\hat{f}(k)$ and the corresponding values of $\hat{E}(k)$ for $k = 45$ and $k = 50$. [6]

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